ANABIONTICS

The Navigation of Biological State

Essay II

Hiram Dunn
AstroBiota
14 August 2026

1. From Persistence to Accessibility

The first theory of anabiontics begins with persistence. Living systems do not simply occupy environments; they continuously construct, inherit, and modify the material conditions through which their organization persists. Molecules, polymers, extracellular matrices, substrates, interfaces, and constructed environments can therefore participate in biological continuity without themselves needing to remain permanently present. Anabiontics provides a language for these materials according to their role in biological state rather than according to chemical identity alone.

There is, however, a larger consequence of this theory. If material conditions participate in determining which biological states persist, they must also participate in determining which biological states are accessible. Anabiontics therefore cannot be limited to the maintenance of states that already exist. It must also address the movement of living systems through biological state space.

This changes the meaning of biological possibility. Biology normally describes organisms according to states we have already observed. A fungus has characteristic morphologies. A bacterium has recognizable metabolic and developmental programs. A tissue has familiar architectures. A microbial community tends toward certain compositions. We naturally begin treating these observations as descriptions of what the biological system is capable of doing. But an observed phenotype represents only a state that a particular biological system reached under a particular history of constraints.

The observed organism may represent only a fraction of the possible organism.

A genome establishes important boundaries on biological possibility, but it does not necessarily specify a single biological state. The state that emerges depends on interactions among genetic capacity, previous biological history, material environment, neighboring organisms, developmental timing, energetic conditions, and physical constraint. Two genetically equivalent systems can consequently occupy meaningfully different states, and identical environments need not produce identical outcomes when the systems entering those environments carry different histories.

This introduces a distinction between possible state and accessible state. A biological state may be physically and genetically possible without being directly accessible from the system's present organization. Humans may therefore have observed only a small subset of the states available to many biological systems, not because the remaining states require new genomes, but because we have never supplied the histories necessary to reach them.

Anabiontics offers a material theory for navigating this problem.

2. State Access and State Maintenance

An anabiontic does not need to contain the state that follows from its introduction. It may instead change the constraints sufficiently for the system to enter a state that was previously inaccessible. The material becomes significant because it changes the trajectory available to biology. Once that trajectory has been entered, subsequent biological processes may establish their own architecture of persistence, potentially allowing the initiating material to disappear.

This separates state access from state maintenance. A material may open access to a state without continuously maintaining it. Another material may stabilize that state. Still another may participate in reconstructing it after disturbance. These functions may also occur within the same material at different stages of the transition. Anabiontic classification is therefore inherently dependent on time, history, and the state of the receiving system.

This also means that biological state transitions cannot always be reduced to a relationship between one material and one outcome. Biological trajectories may be ordered. If a system can occupy states A, B, C, and D, it does not follow that A can transition directly to D. State B may construct the conditions necessary for C, while C may construct conditions necessary for D. Under those circumstances, the final state cannot be produced merely by reproducing the final environment. The history that generated that environment is itself part of the causal architecture.

The order of material exposure can therefore matter. Exposure to X followed by Y may produce a different state than exposure to Y followed by X because the second material never encounters the original biological system. It encounters the state produced by the first exposure. Even when the chemical inputs are identical, the biological events are not.

This suggests that anabiontic engineering may eventually involve the design of material histories rather than individual materials. Conventional screening frequently asks what happens when a biological system encounters compound X. Anabiontic screening can additionally ask what becomes possible after X, what subsequently becomes possible after Y, what persists after both disappear, and whether reversing their order produces a different destination. Duration, sequence, withdrawal, repetition, starting state, and recovery therefore become fundamental experimental variables.

History becomes experimentally engineerable.

3. State Omega

This framework permits a more ambitious objective than restoration. Consider a biological system in State A and a desirable state, Omega, that the system has never previously occupied. Omega does not necessarily require genetic modification. It may already exist within the physical and biological possibility space of the system while remaining inaccessible from A under the material histories it normally experiences.

The objective of anabiontic engineering would be to construct a trajectory from A to Omega.

The distinction from conventional restoration is important. There is no previous Omega to restore. The system has never occupied it. An intervention that provides access to Omega is therefore not returning biology to an earlier condition. It is exposing a biological possibility that had previously remained inaccessible.

The first question is whether Omega can be established. The more important question is what happens when the material environment used to establish it disappears.

If Omega immediately collapses when the initiating material is withdrawn, then the state was primarily being maintained externally. That may still be useful, but it represents a relatively weak form of state engineering. If instead biological processes initiated during the transition begin maintaining the new organization themselves, persistence has partially transferred from the intervention to the biological system.

The strongest case would be reconstruction. Living systems cannot preserve state by permanently retaining their components. Cells die and are replaced. Proteins and lipids turn over. Extracellular structures are remodeled. Microbial populations fluctuate. Metabolites disappear. Nevertheless, biological organization can remain recognizable over periods far longer than the lifetimes of many of its individual components.

Persistence must therefore involve more than material permanence. It involves the ability to reconstruct organization through material turnover.

A meaningful State Omega would consequently not have to remain perfectly unchanged. It would have to become a state toward which the system preferentially reorganizes. After ordinary disturbance, component replacement, or environmental fluctuation, the system would tend to reconstruct Omega rather than simply collapse toward its historical State A.

Biological persistence may be better measured through recovery than through immobility.

The question is not merely whether a state survives disturbance unchanged. The more revealing question is what state the system constructs after disturbance has displaced it.

Anabiontics therefore concerns attractors as much as structures. A sufficiently established biological state becomes important not because every molecule remains where it was, but because the coupled system has acquired a tendency to rebuild a recognizable organization.

4. Anabiontic Recall

A further possibility follows from this framework. Suppose material X participates in establishing Omega and is subsequently removed. The biological system then experiences sufficient disturbance to move away from Omega. If X is introduced again, its second encounter occurs in a system with a fundamentally different history from the first.

The material has not necessarily changed. The receiving biology has.

If the second encounter with X preferentially facilitates reconstruction of Omega, then the biological meaning of X has become history-dependent. The first encounter participated in constructing a trajectory; the later encounter occurs in a system carrying traces of that trajectory. I describe this proposed phenomenon as anabiontic recall.

This does not require the material to contain a symbolic representation of the previous state. Biological memory does not need to operate symbolically. Previous state can alter cellular composition, regulatory architecture, extracellular structure, immune populations, microbial organization, spatial relationships, or other persistence substrates. The consequence is simply that the same material can produce a different biological event because the system encountering it has changed.

This is an important distinction between ordinary repeated pharmacology and anabiontic recall. If X produces exactly the same effect every time because of the same immediate mechanism, nothing unusual needs to be invoked. The stronger phenomenon occurs when prior history with X changes the later response to X and biases reconstruction toward a previously established state.

Biological history has then become part of the material response.

5. State Delta

Nothing in this theory requires newly accessible or persistent states to be beneficial. Biological persistence has no intrinsic preference for health. Pathological states can stabilize themselves. Inflammatory states can acquire memory. Ecological configurations can become resistant to restoration. A system can become better at reconstructing an undesirable state just as it might become better at reconstructing a desirable one.

I designate this undesirable region of state space State Delta.

The distinction between Omega and Delta is critical because it prevents anabiontics from becoming a theory in which persistence is automatically interpreted as biological improvement. An anabiontic transition has a direction, but its value depends on the resulting state and the system in which that state occurs.

Allergic sensitization provides an especially useful model for understanding this problem. A transient material encounter can alter an immune system such that later exposure to the same material produces a substantially different response. The material does not need to remain continuously present between exposures. What persists is a changed biological architecture capable of responding differently when the material returns.

This is particularly relevant to lichen-derived materials. Some lichen metabolites are capable of producing contact sensitization in susceptible individuals. Within conventional immunology, the mechanisms of sensitization and subsequent inflammatory responses are already studied without requiring anabiontic terminology. Anabiontics contributes something different: it places that event within the broader problem of state trajectory. The first exposure and later exposure cannot be treated as equivalent events because the first can change the state of the system that receives the second.

In the language proposed here, sensitization represents movement toward Delta. A material encounter has produced biological history that changes the consequences of future encounters. That is precisely why anabiontic engineering cannot be reduced to finding materials capable of producing persistent effects. The central problem is determining where in state space those effects are taking the system.

The objective is not persistence alone. It is directed persistence.

6. From Material Intervention to State Engineering

Anabiontics therefore expands the target of biological engineering. Modern biotechnology has developed extraordinary tools for modifying components: genes can be edited, receptors activated or inhibited, organisms eliminated, cells transplanted, proteins replaced, and metabolic pathways manipulated. These interventions remain indispensable, but they do not exhaust the possibilities of biological engineering.

A state-centered biotechnology asks a different question. Given a desired biological organization, what sequence of material constraints makes that organization accessible, what allows it to establish, what transfers persistence into the biological system, and what allows the system to reconstruct that organization after disturbance?

This question applies across scale because state itself is not restricted to a particular level of biological organization. Cells occupy states, but so do tissues. Microbial communities occupy states. Symbioses occupy states. Organisms and constructed ecologies occupy states. Moreover, a material acting at one scale can alter state at another. A small molecule can initiate tissue remodeling. A polymer can reorganize cellular interfaces. An extracellular matrix can alter differentiation. A microorganism can alter the state of another organism. A community can construct a material environment that changes the future behavior of every organism inhabiting it.

Anabiontics therefore should not be understood simply as a new branch of pharmacology. Pharmacology is one territory within a much larger material landscape. The relevant anabiontic may be a molecule, polymer, matrix, biological secretion, conditioned substrate, organism-generated material environment, or sequence of several of these. What unifies them is not chemistry but their role in determining the accessibility and persistence of biological state.

This leads to a different conception of biological technology. The intervention itself is not necessarily the final product. A molecule may disappear. A polymer may degrade. An administered organism may vanish. A constructed environment may be temporary. If the intervention has nevertheless changed which state the biological system can establish or reconstruct, then the enduring product is the altered biological state.

That possibility also changes how we should think about biological diversity. Some of the diversity available to life may not be taxonomic. It may be state diversity. Organisms whose genomes are already known may possess stable phenotypes, coupled organizations, metabolic configurations, developmental trajectories, or ecological relationships that humans have never observed because we have repeatedly examined them within a narrow range of material histories.

Anabiontics therefore proposes a new experimental frontier: rather than searching only for new organisms, genes, or molecules, biology can systematically search for new states of systems we already possess.

The central question becomes whether there are regions of biological state space that are physically possible but historically inaccessible. If so, accessing them will require more than identifying their components. It will require discovering the sequence of constraints through which those states can be constructed.

This is the larger consequence of anabiontics. The theory begins with the materials through which biological state persists, but it leads inevitably to the materials through which biological possibility becomes accessible.

The frontier is no longer only what life is made of.

It is where life can go.